M12x Unit Catalogue

Graphs, mechanisms, data tables, and intended learned rules

Expanded Milestone 1 Part 2 (units U1–U7)

Causal ABA Learning project — Samuel Waugh

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Abstract

This document catalogues the seven experimental *units* used in the expanded Milestone 1 Part 2 grid (experiment identifier M12x). Each unit specifies a directed acyclic graph, a data-generating mechanism, a finite observational table, and one or more learning targets together with an intended learned rule for that target. The catalogue records *inputs* to ABA Learning only; it does not report learner outputs. It is written so that a reader unfamiliar with the project shorthand can understand what is held fixed, what varies across units, and what rule each cell is designed to probe.

Contents

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1 Purpose of this catalogue

The M12x experiment asks when and how two published ABA Learning configurations (ECAI 2024 and AAMAS 2025) recover intended intensional rules from categorical tabular data generated by known mechanisms on small directed acyclic graphs. The experiment uses the project's existing ABA Learning bridge over tabular data. It does *not* implement Russo-style Causal ABA (arrow, no-edge, or independence assumptions with stable extensions as candidate graphs).

A **unit** is one graph together with one data-generating process and a fixed set of learning targets. A **cell** is one triple consisting of a unit, a learning target, and a learner configuration. There are seven units and ten unit–target pairs. Crossing each pair with the two configurations yields twenty cells in total.

For each unit, the sections below give:

1. a natural-language statement of the unit's scientific purpose;
2. the graph;
3. the mechanisms used to generate the table;
4. the observational table $\mathcal{D}$;
5. for each learning target, the positive and negative example sets and the intended learned rule $\mathcal{H}^\star_t$.

2 Notation and terminology

Graph. Let $G = (V, E)$ be a directed acyclic graph. The node set is written $V = \{x_0, x_1, \ldots\}$ and the edge set $E \subseteq V \times V$. An edge $x_i \to x_j$ means that x_i is a direct cause of x_j in the data-generating story for that unit. The **parents** of a node x are $\mathrm{Pa}(x) = \{y \in V : (y \to x) \in E\}$. The **ancestors** of x are all nodes with a directed path to x (excluding x). The **descendants** of x are all nodes reachable from x by a directed path (excluding x). The **sources** are $S = \{x \in V : \mathrm{Pa}(x) = \emptyset\}$. The **non-sources** are $N = V \setminus S$.

Alphabet and table. Every variable takes values in the finite alphabet $K = \{0, 1, 2\}$. An observational table $\mathcal{D}$ is a finite list of rows. Each row is an assignment of a value in K to every variable in V. Rows are indexed by **sample identifiers** $1, 2, \ldots, |\mathcal{D}|$ in the order shown in each unit's table.

Mechanisms. Except where a unit explicitly uses curated multi-row support (U5 and U7), the table is built by taking the full Cartesian product $K^{|S|}$ over source assignments and then evaluating a deterministic function $x := f_x(\mathrm{Pa}(x))$ for each non-source x in a topological order. The notation $\mathrm{copy}(y)$ means the identity map $y \mapsto y$. The notation $|a - b|$ means absolute difference on the integers $\{0, 1, 2\}$.

Learning target and examples. A **learning target** $t \in N$ is the variable whose predicate the learner must characterise. Under the shared labelling regime used for all units in this catalogue, the **positive examples** are

$$E^+ = \{\, i \in \{1, \ldots, |\mathcal{D}|\} : \text{the value of } t \text{ in row } i \text{ is not } 0 \,\},$$

and the **negative examples** are

$$E^- = \{\, i \in \{1, \ldots, |\mathcal{D}|\} : \text{the value of } t \text{ in row } i \text{ is } 0 \,\}.$$

In the ABA Learning input these appear as ground atoms $t(i)$ for $i \in E^+$ (positive) and $i \in E^-$ (negative).

Background knowledge. For a fixed target t, the background knowledge contains exact-value predicates for every variable other than t. The atom $x_i\_\mathtt{val}\_v(A)$ is true of sample identifier A precisely when variable x_i equals v in that row. If t has descendants in G, those descendants remain in the background knowledge as available predictors. Citing a descendant in a learned rule for t is recorded as an explicit failure mode in later inspection.

Intended learned rule. For each target t, the symbol $\mathcal{H}_t^\star$ denotes a compact set of Horn clauses in the `val` language that correctly separates E^+ from E^- on $\mathcal{D}$ and that uses the intended parent (or parents) of t. This is the reference against which learned rules are compared. It is an intensional design target for inspection, not a claim that the learner must emit that exact syntax.

Configurations. The two learner configurations are the published Prolog files abbreviated ECAI and AAMAS. Their filenames are `ecai2024_config.pl` and `aamas2025_config.pl`. Both files live in the repository directory named `configs`.

3 Shared experimental regime

The following choices are held fixed across U1–U7 unless a unit section states an explicit exception.

1. Every variable has alphabet $K = \{0, 1, 2\}$.
2. Labels are nonzero-positive: E^+ contains exactly the rows where $t \neq 0$, and E^- contains exactly the rows where $t = 0$.
3. Background knowledge uses only `val` predicates (exact-value atoms $x_i\_\mathtt{val}\_v$).
4. Units U1–U4 and U6 build $\mathcal{D}$ by full source factorial over $K^{|S|}$ followed by deterministic propagation. Units U5 and U7 instead use a curated multi-row support that is not a single-valued function of the sources alone; those constructions are stated in full in their sections.
5. Each unit–target pair is run once under ECAI and once under AAMAS, giving twenty cells in total.

Overview of units

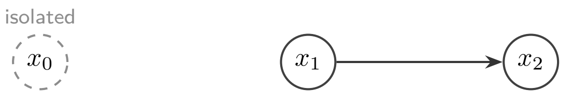
ID	Graph family	Learning targets	$\lvert S\rvert$	$\lvert\mathcal{D}\rvert$	Cells
U1	separator with isolated x_0	$\{x_2\}$	2	9	2
U2	collider	$\{x_2\}$	2	9	2
U3	collider	$\{x_2\}$	2	9	2
U4	fork	$\{x_1, x_2\}$	1	3	4
U5	chain (curated support)	$\{x_1, x_2\}$	1	6	4
U6	four-node G1	$\{x_2, x_3\}$	2	9	4
U7	diamond (curated support)	$\{x_3\}$	1	6	2
		10 targets			**20**

4 U1 — Separator with copy

Purpose. Unit U1 tests whether ABA Learning recovers a unary parent rule for a sink that is an exact copy of its unique parent, while an isolated irrelevant variable remains available in the background knowledge. The intended behaviour is to cite the true parent and not the isolated distractor.

Fixture key: `m12_u1_separator_copy`. **Learning targets:** $\{x_2\}$.

4.1 Graph



The node set is $V = \{x_0, x_1, x_2\}$. The edge set is $E = \{x_1 \to x_2\}$. The sources are $S = \{x_0, x_1\}$ and the non-sources are $N = \{x_2\}$. Variable x_0 has no incident edges.

4.2 Mechanisms

The only non-source mechanism is

$$x_2 \ := \ \text{copy}(x_1).$$

Variable x_0 does not enter any mechanism.

4.3 Observational table

The table is the full factorial over $(x_0, x_1) \in K \times K$, after which x_2 is set equal to x_1. There are nine rows.

id	x_0	x_1	x_2
1	0	0	0
2	0	1	1
3	0	2	2
4	1	0	0
5	1	1	1
6	1	2	2
7	2	0	0
8	2	1	1
9	2	2	2

4.4 Target $t = x_2$

Learning target x_2

The positive examples are the six rows where $x_2 \neq 0$, namely $E^+ = \{2, 3, 5, 6, 8, 9\}$. The negative examples are $E^- = \{1, 4, 7\}$. Background knowledge includes `val` predicates for x_0 and x_1 and excludes x_2. The ancestors of x_2 are $\{x_1\}$. The isolated variable x_0 is a distractor in the background knowledge. There are no descendants of x_2.

The intended learned rule is

```
x2(A) :- x1_val_1(A).
x2(A) :- x1_val_2(A).
```

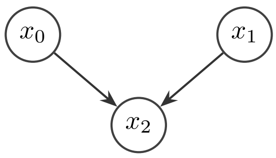
These clauses characterise $x_2 \neq 0$ by the parent x_1 being nonzero. They must not cite the isolated distractor x_0.

5 U2 — Collider with minimum

Purpose. Unit U2 tests recovery of a conjunctive both-parent rule. The sink is the minimum of two independent parents, so under nonzero-positive labelling the target is nonzero if and only if both parents are nonzero.

Fixture key: `m12_u2_collider_min`. **Learning targets:** $\{x_2\}$.

5.1 Graph



The node set is $V = \{x_0, x_1, x_2\}$. The edge set is $E = \{x_0 \to x_2,\ x_1 \to x_2\}$. The sources are $S = \{x_0, x_1\}$ and the non-sources are $N = \{x_2\}$. The parents of x_2 are $\mathrm{Pa}(x_2) = \{x_0, x_1\}$.

5.2 Mechanisms

$$x_2 \ := \ \min(x_0, x_1).$$

5.3 Observational table

The table is the full factorial over $(x_0, x_1) \in K \times K$. There are nine rows.

id	x_0	x_1	$x_2 = \min(x_0, x_1)$
1	0	0	0
2	0	1	0
3	0	2	0
4	1	0	0
5	1	1	1
6	1	2	1
7	2	0	0
8	2	1	1
9	2	2	2

5.4 Target $t = x_2$

Learning target x_2

The positive examples are $E^+ = \{5, 6, 8, 9\}$ and the negative examples are $E^- = \{1, 2, 3, 4, 7\}$. On this table, $x_2 \neq 0$ if and only if $x_0 \neq 0$ and $x_1 \neq 0$. Background knowledge includes `val` predicates for x_0 and x_1.

The intended learned rule is

```
x2(A) :- x0_val_1(A), x1_val_1(A).
x2(A) :- x0_val_1(A), x1_val_2(A).
x2(A) :- x0_val_2(A), x1_val_1(A).
x2(A) :- x0_val_2(A), x1_val_2(A).
```

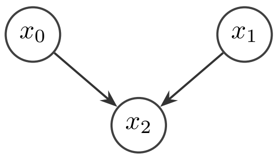
These four clauses are the `val` expansion of "both parents are nonzero".

6 U3 — Collider with maximum

Purpose. Unit U3 is the disjunctive counterpart of U2. The sink is the maximum of the same two parents, so under nonzero-positive labelling the target is nonzero if and only if at least one parent is nonzero. The intended learned rule is a two-parent disjunction expressed as four single-literal clauses.

Fixture key: `m12_u3_collider_max`. **Learning targets:** $\{x_2\}$.

6.1 Graph



The graph is identical to U2: $V = \{x_0, x_1, x_2\}$, $E = \{x_0 \to x_2,\ x_1 \to x_2\}$, $S = \{x_0, x_1\}$, and $N = \{x_2\}$.

6.2 Mechanisms

$$x_2 \ := \ \max(x_0, x_1).$$

6.3 Observational table

id	x_0	x_1	$x_2 = \max(x_0, x_1)$
1	0	0	0
2	0	1	1
3	0	2	2
4	1	0	1
5	1	1	1
6	1	2	2
7	2	0	2
8	2	1	2
9	2	2	2

6.4 Target $t = x_2$

Learning target x_2

The positive examples are $E^+ = \{2, 3, 4, 5, 6, 7, 8, 9\}$ and the negative example set is $E^- = \{1\}$. On this table, $x_2 \neq 0$ if and only if $x_0 \neq 0$ or $x_1 \neq 0$. Background knowledge includes `val` predicates for x_0 and x_1.

The intended learned rule is

```
x2(A) :- x0_val_1(A).
x2(A) :- x0_val_2(A).
x2(A) :- x1_val_1(A).
x2(A) :- x1_val_2(A).
```

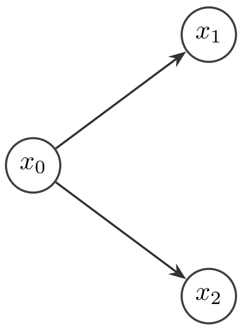
These clauses are the `val` expansion of "at least one parent is nonzero".

7 U4 — Fork with asymmetric maps

Purpose. Unit U4 tests parent recovery under a fork when a correlated sibling is present in the background knowledge. Each child is a deterministic but asymmetric function of the common cause x_0, so the true parent separates the nonzero labelling of that child while the sibling is associated with the target yet imperfect as a separator.

Fixture key: `m12_u4_fork_double_copy`. **Learning targets:** $\{x_1, x_2\}$.

7.1 Graph



The node set is $V = \{x_0, x_1, x_2\}$. The edge set is $E = \{x_0 \to x_1, x_0 \to x_2\}$. The source set is $S = \{x_0\}$ and the non-source set is $N = \{x_1, x_2\}$.

7.2 Mechanisms

$$x_1 := \begin{cases} 2 & \text{if } x_0 \neq 2, \\ 0 & \text{if } x_0 = 2, \end{cases}$$

$$x_2 := \begin{cases} 2 & \text{if } x_0 \neq 0, \\ 0 & \text{if } x_0 = 0. \end{cases}$$

7.3 Observational table

The table is the full factorial over $x_0 \in K$, which yields three rows.

id	x_0	x_1	x_2
1	0	2	0
2	1	2	2
3	2	0	2

7.4 Target $t = x_1$

Learning target x_1

The positive examples are $E^+ = \{1, 2\}$ and the negative examples are $E^- = \{3\}$. On this table, $x_1 \neq 0$ if and only if $x_0 \in \{0, 1\}$. Background knowledge includes `val` predicates for the parent x_0 and the sibling x_2. Citing the sibling is a divergence.

The intended learned rule is

```
x1(A) :- x0_val_0(A).
x1(A) :- x0_val_1(A).
```

7.5 Target $t = x_2$

Learning target x_2

The positive examples are $E^+ = \{2, 3\}$ and the negative examples are $E^- = \{1\}$. On this table, $x_2 \neq 0$ if and only if $x_0 \neq 0$. Background knowledge includes `val` predicates for the parent x_0 and the sibling x_1. Citing the sibling is a divergence.

The intended learned rule is

```
x2(A) :- x0_val_1(A).
x2(A) :- x0_val_2(A).
```

8 U5 — Chain with curated correlated ancestor

Purpose. Unit U5 restores the pilot chain construction in which the intermediate is not a single-valued function of the root. For the sink x_2, the unit tests whether the learner prefers the direct parent x_1 over a correlated but imperfect ancestor x_0. For the intermediate x_1, the descendant x_2 remains in the background knowledge; on this table that descendant is extensionally tied to x_1, so citing it is an explicit failure mode rather than an intended rule.

Fixture key: `m12_u5_chain_double_copy`. **Learning targets:** $\{x_1, x_2\}$.

8.1 Graph

$$x_0 \longrightarrow x_1 \longrightarrow x_2$$

The node set is $V = \{x_0, x_1, x_2\}$. The edge set is $E = \{x_0 \to x_1,\ x_1 \to x_2\}$. The source set is $S = \{x_0\}$ and the non-source set is $N = \{x_1, x_2\}$.

8.2 Mechanisms

This unit does *not* use full source factorial with a single-valued map $x_1 = f(x_0)$. Instead, for each $x_0 \in K$ the table includes the two pairs $(x_0, x_1) = (x_0, x_0)$ and $(x_0, (x_0 + 1) \bmod 3)$. The sink is then deterministic:

$$x_2 \ := \ \mathrm{copy}(x_1).$$

Thus x_1 is multi-valued given x_0, while x_2 equals x_1 in every row.

8.3 Observational table

There are six rows.

id	x_0	x_1	x_2
1	0	0	0
2	0	1	1
3	1	1	1
4	1	2	2
5	2	2	2
6	2	0	0

8.4 Target $t = x_1$

Learning target x_1

The positive examples are $E^+ = \{2, 3, 4, 5\}$ and the negative examples are $E^- = \{1, 6\}$. Because $x_2 = x_1$ on every row, $x_1 \neq 0$ if and only if $x_2 \neq 0$. The parent x_0 is an imperfect separator of this labelling: each of $x_0 = 0$ and $x_0 = 2$ appears on both a positive and a negative row. Background knowledge includes `val` predicates for x_0 and for the descendant x_2.

There is *no* parent-aligned intended rule $\mathcal{H}^\star_{x_1}$. No set of clauses using only `x0_val_`$*$ separates $E^\pm$ perfectly. Rules that cite only the descendant x_2 are extensionally perfect on $\mathcal{D}$ but are treated as the descendant-citation failure mode, not as the reference.

8.5 Target $t = x_2$

Learning target x_2

The positive examples are $E^+ = \{2, 3, 4, 5\}$ and the negative examples are $E^- = \{1, 6\}$. On this table, $x_2 \neq 0$ if and only if $x_1 \in \{1, 2\}$. The ancestor x_0 is imperfect for the same reason as above. Background knowledge includes `val` predicates for x_0 and x_1.

The intended learned rule uses the direct parent:

```
x2(A) :- x1_val_1(A).
x2(A) :- x1_val_2(A).
```

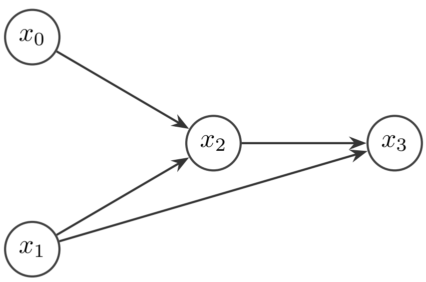
A rule set that uses only the ancestor x_0 cannot match this labelling perfectly. Ancestor citation when the parent is available is a divergence from $\mathcal{H}^\star_{x_2}$.

9 U6 — Four-node G1 with minimum then difference

Purpose. Unit U6 uses the four-node Fabrizio G1 edge set with project-chosen mechanisms. The intermediate x_2 is a minimum of two sources, so its nonzero labelling is the same conjunctive concept as in U2, now with a descendant in the background knowledge. The sink x_3 is the arithmetic difference $x_1 - x_2$; under nonzero-positive labelling this is exactly the strict inequality $x_1 > x_2$, which requires both parents of x_3.

Fixture key: `m12_u6_g1_and_cone`. **Learning targets:** $\{x_2, x_3\}$.

9.1 Graph



The node set is $V = \{x_0, x_1, x_2, x_3\}$. The edge set is $E = \{x_0 \to x_2,\ x_1 \to x_2,\ x_1 \to x_3,\ x_2 \to x_3\}$. The sources are $S = \{x_0, x_1\}$ and the non-sources are $N = \{x_2, x_3\}$. The parent sets are $\mathrm{Pa}(x_2) = \{x_0, x_1\}$ and $\mathrm{Pa}(x_3) = \{x_1, x_2\}$.

9.2 Mechanisms

$$
\begin{aligned}
x_2 &:= \min(x_0, x_1), \\
x_3 &:= x_1 - x_2.
\end{aligned}
$$

Because $x_2 \le x_1$ holds on every row of the factorial table, the difference x_3 always lies in $\{0, 1, 2\}$.

9.3 Observational table

id	x_0	x_1	$x_2 = \min(x_0, x_1)$	$x_3 = x_1 - x_2$
1	0	0	0	0
2	0	1	0	1
3	0	2	0	2
4	1	0	0	0
5	1	1	1	0
6	1	2	1	1
7	2	0	0	0
8	2	1	1	0
9	2	2	2	0

9.4 Target $t = x_2$

Learning target x_2

The positive examples are $E^+ = \{5, 6, 8, 9\}$ and the negative examples are $E^- = \{1, 2, 3, 4, 7\}$. On this table, $x_2 \neq 0$ if and only if $x_0 \neq 0$ and $x_1 \neq 0$. The descendant x_3 is nonzero only on rows $\{2, 3, 6\}$, so it is an imperfect separator of $E^\pm$ for x_2. Background knowledge includes `val` predicates for x_0, x_1, and x_3.

The intended learned rule is the same conjunctive pattern as in U2:

```
x2(A) :- x0_val_1(A), x1_val_1(A).
x2(A) :- x0_val_1(A), x1_val_2(A).
x2(A) :- x0_val_2(A), x1_val_1(A).
x2(A) :- x0_val_2(A), x1_val_2(A).
```

These clauses must not cite the descendant x_3.

9.5 Target $t = x_3$

Learning target x_3

The positive examples are $E^+ = \{2, 3, 6\}$ and the negative examples are $E^- = \{1, 4, 5, 7, 8, 9\}$. On this table, $x_3 \neq 0$ if and only if $x_1 > x_2$. Neither parent alone separates this labelling. Background knowledge includes `val` predicates for x_0, x_1, and x_2. The variable x_0 is a non-parent ancestor of x_3.

The intended learned rule is the parent-based case split of $x_1 > x_2$:

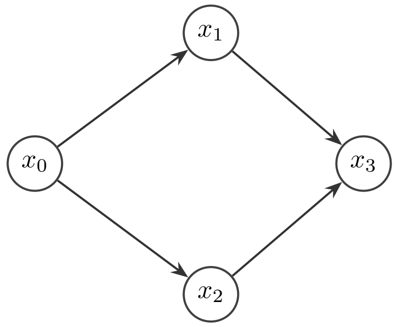
```
x3(A) :- x1_val_1(A), x2_val_0(A).
x3(A) :- x1_val_2(A), x2_val_0(A).
x3(A) :- x1_val_2(A), x2_val_1(A).
```

10 U7 — Diamond with curated noisy fork arms

Purpose. Unit U7 tests both-parent recovery for a sink whose parents are correlated siblings under a common cause. A three-row deterministic factorial diamond would allow a perfect root-only case-split of many labellings. To prevent that escape, U7 uses a curated support in the style of U5: for each root value, two sibling configurations are included so that the same x_0 appears on both a positive and a negative row for the sink labelling. The intended rule therefore cannot be replaced by a perfect `x0_val_*` union, and each sibling alone is likewise imperfect.

Fixture key: `m12_u7_g1_or_cone` (historical key retained for path stability). **Learning targets:** $\{x_3\}$.

10.1 Graph



The node set is $V = \{x_0, x_1, x_2, x_3\}$. The edge set is $E = \{x_0 \to x_1,\ x_0 \to x_2,\ x_1 \to x_3,\ x_2 \to x_3\}$. The source set is $S = \{x_0\}$. The non-sources are $N = \{x_1, x_2, x_3\}$, of which only x_3 is a learning target. The parents of the sink are $\mathrm{Pa}(x_3) = \{x_1, x_2\}$.

10.2 Mechanisms

For each $x_0 \in K$, set $x_1 := (x_0 + 1) \bmod 3$ and include two values of x_2:

1. $x_2 = (x_0 + 2) \bmod 3$ (the siblings differ);
2. $x_2 = x_1$ (the siblings collapse).

The sink is then

$$x_3 \ := \ |x_1 - x_2|.$$

On the resulting table, $x_3 \neq 0$ if and only if $x_1 \neq x_2$.

10.3 Observational table

There are six rows.

id	x_0	x_1	x_2	$x_3 = \|x_1 - x_2\|$
1	0	1	2	1
2	0	1	1	0
3	1	2	0	2
4	1	2	2	0
5	2	0	1	1
6	2	0	0	0

10.4 Target $t = x_3$

Learning target x_3

The positive examples are $E^+ = \{1, 3, 5\}$ and the negative examples are $E^- = \{2, 4, 6\}$. On this table, $x_3 \neq 0$ if and only if $x_1 \neq x_2$. No `val`-union over $\{x_0\}$ alone, over $\{x_1\}$ alone, or over $\{x_2\}$ alone separates $E^\pm$ perfectly. The pair $\{x_1, x_2\}$ does separate $E^\pm$. Background knowledge includes `val` predicates for x_0, x_1, and x_2.

The intended learned rule is

```
x3(A) :- x1_val_1(A), x2_val_2(A).
x3(A) :- x1_val_2(A), x2_val_0(A).
x3(A) :- x1_val_0(A), x2_val_1(A).
```

These three clauses list the sibling value pairs on which the siblings differ.

Source pointers

Mechanism cards
`docs/research/milestone_plans/milestone1_part2/mechanism_cards/`

Unit set
`milestone1_part2_expanded_unit_set.md`

Approach
`milestone1_part2_expanded_approach.md`

Implementation
`causal/experiments/handcrafted_m12x.py`